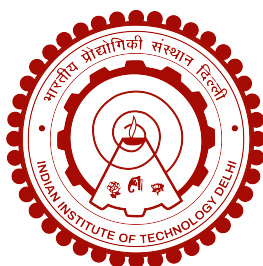


Stochastic Reaction Network Modeling and Optimal Control of Bacterial Populations Under Antibiotic Stress

Investigating Antibiotic Resistance Emergence under Entropic Control Theory

A project proposal submitted for the consideration of the

Summer Undergraduate Research Award (SURA)



Facilitator:

Garima Rani

Assistant Professor

Department of Biochemical Engineering and Biotechnology

Indian Institute of Technology Delhi

Head of Department (Biochemical Engineering and Biotechnology):

Preeti Srivastava

Professor

Department of Biochemical Engineering and Biotechnology

Indian Institute of Technology Delhi

Head of Department (Mathematics):

Mani Mehra

Professor

Department of Mathematics

Indian Institute of Technology Delhi

Applicants:

Tanay Singh Lodhi

2024MT60631

CGPA: 8.61

mt6240631@iitd.ac.in

Contact no: 7042480861

Monil Savaliya

2024MT60344

CGPA: 9.05

mt6240344@iitd.ac.in

Contact no: 9328552413

Abstract

Antibiotic resistance is one of the most critical public health challenges of the 21st century, responsible for over a million deaths annually. The dynamics of resistance emergence are fundamentally stochastic: a single resistant mutant arising by chance in a bacterial population can escape extinction and eventually dominate, making deterministic population models inadequate for capturing such rare but catastrophic transitions.

This project investigates how stochastic reaction network (SRN) models can be used to simulate and control bacterial population dynamics under antibiotic stress, with particular focus on rare events such as resistance emergence. We build on the theoretical framework of Horiguchi and Kobayashi [1], which established optimal control for stochastic reaction networks using Kullback–Leibler divergence as a control cost. Specifically, we will incorporate time-varying natural dynamics to reflect realistic drug concentration profiles and investigate how control strategies influence rare stochastic events such as resistance emergence—both addressing critical gaps in the current framework. Additionally, we will develop approaches for scalable simulation and parameter inference, and speculatively explore actor–critic reinforcement learning as a complementary policy learning approach, contingent on project progress. The core expected outcome is a computational framework for adaptive antibiotic dosing that accounts for the time-varying nature of drug concentration and explicitly quantifies the stochastic risk of resistance emergence. Extended outcomes include scalable simulation methods for large populations, parameter inference frameworks, and preliminary RL-based dosing policies.

Introduction

Bacterial populations under antibiotic stress evolve through discrete stochastic events — cell birth, death, and mutation — governed by reaction rates that depend on both the internal population state and external conditions such as drug concentration. A key feature of this system is that antibiotic concentration is not constant during treatment. Drug levels rise after each dose and decay between doses, following pharmacokinetic profiles that vary with dosing schedule, patient physiology, and the antibiotic in question. This means that the natural death rate of sensitive bacteria is itself a time-varying quantity, changing continuously throughout the course of treatment. Any realistic model of antibiotic control must account for this inherently time-dependent nature of the biological environment.

Beyond time-varying dynamics, resistance emergence presents a second fundamental challenge. Resistance is a rare event: in most treatment scenarios, resistant mutants arise infrequently and are often eliminated by chance before establishing. But when they do establish, the consequences are severe. This asymmetry — rare occurrence, catastrophic outcome — motivates studying control strategies that are specifically designed to guard against such events, rather than strategies optimized only for typical population trajectories. Understanding how different dosing schedules influence the probability of resistance emergence, and not just average population counts, is therefore a central question. The population-level consequences of such dynamics have been studied extensively in classical epidemiological frameworks [3], and path-integral methods have provided complementary tools for stochastic control [4].

Stochastic reaction network (SRN) models, formulated as continuous-time Markov jump processes, provide the natural framework for this problem [2]. The population state evolves through

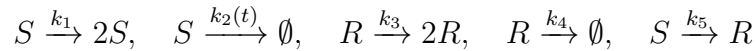
random reaction events, and the full probability distribution over states satisfies the Chemical Master Equation. Recent work by Horiguchi and Kobayashi [1] developed a rigorous optimal control theory for such systems using information-theoretic control costs. A key result of this work is that choosing the Kullback–Leibler divergence as a control cost renders the otherwise intractable Hamilton–Jacobi–Bellman equation exactly solvable, and that optimal strategies exhibit a striking mode-switching structure: the controller naturally alternates between active intervention and passive phases depending on the population state. This is directly analogous to real treatment protocols that alternate between dosing and rest periods.

However, critical gaps remain before this framework can be applied to antibiotic resistance in realistic settings. First, the theory assumes fixed reaction rates, whereas drug concentration—and hence killing rates—changes continuously during treatment. Second, the framework has not been examined specifically for controlling rare adverse events like resistance emergence, which is clinically the most critical objective. This project addresses these central gaps through theoretical extension and rare event analysis. Additionally, we will develop approaches toward scalable simulation for large populations and parameter inference from noisy observations, and speculatively investigate whether reinforcement learning [5] offers a tractable complementary approach for adaptive dosing in this stochastic setting, extending the framework toward fully data-driven deployment.

Objectives

The objectives of this project are structured as follows, prioritizing core contributions to the field:

1. To study and implement the optimal control framework for stochastic reaction networks developed in [1], and verify its key results computationally for benchmark systems before extending it to bacterial population models.
2. To construct a stochastic reaction network model of bacterial populations under antibiotic stress, incorporating sensitive bacteria S , resistant bacteria R , and the following elementary reactions:



where k_1 is the birth rate of sensitive bacteria, $k_2(t)$ is the time-varying antibiotic-induced killing rate of sensitive bacteria, k_3 is the birth rate of resistant bacteria, k_4 is the natural death rate of resistant bacteria, and k_5 is the mutation rate (conversion of sensitive to resistant).

3. To extend the control framework to systems with time-varying natural rates, capturing the pharmacokinetic variation in antibiotic concentration during realistic dosing regimens, and investigate how optimal dosing schedules emerge from this extended framework.
4. To develop scalable Monte Carlo simulation methods for estimating control strategies in large bacterial populations, where direct solution of the master equation is computationally intractable, and to characterize how simulation accuracy scales with the number of sampled trajectories.

In addition to these core objectives, we will explore the following directions pending successful completion of the above:

5. To investigate how model parameters such as bacterial growth and death rates can be inferred from observed population trajectories, and study how parameter uncertainty propagates into the optimal control strategy.
6. To analyze how different control strategies — including those derived from the extended framework — influence the probability of rare events such as resistance emergence, and compare their performance against standard dosing protocols.
7. As a speculative extension, we will explore whether reinforcement learning — specifically actor-critic methods — offers a tractable alternative for learning adaptive dosing policies directly from simulated trajectories. While RL has been applied to antibiotic treatment design [5], its use within stochastic reaction network models with time-varying dynamics and explicit rare-event quantification remains largely unexplored. This direction will be pursued only if the core framework converges with sufficient time remaining.

Approach and Methodology

The project combines theoretical analysis with computational modeling across four phases.

Phase 1 — Foundation (Weeks 1–2). We will study stochastic reaction networks, the Chemical Master Equation, and exact simulation via the Gillespie algorithm [2]. We will implement the computational framework of [1] and reproduce its benchmark results, ensuring correctness before extension. This critical validation step is essential for all downstream work.

Phase 2 — Bacterial Model with Time-Varying Dynamics (Weeks 3–4). We will construct the stochastic reaction network for a two-species bacterial population with growth, death, and mutation reactions. The antibiotic killing rate will be modeled as a time-varying parameter following realistic pharmacokinetic profiles — a significant departure from [1]’s constant-rate assumption. We will characterize baseline resistance emergence probabilities under this time-varying setting, establishing the biological foundation for the control analysis that follows.

Phase 3 — Adaptive Control and Rare Event Analysis (Weeks 5–6). We will extend the control framework to accommodate time-varying rates, computing optimal control strategies by solving the HJB equation at each time step with the current drug concentration. This addresses a critical gap in [1]: mode-switching behavior in response to dynamically changing environments.

Control strategies will be evaluated not by average outcomes (as in [1]) but by their ability to suppress rare events — specifically, resistance emergence. We will compute emergence probabilities via Monte Carlo sampling, construct cost-safety Pareto frontiers, and compare quantitatively against standard constant-dosing protocols. For large populations, scalable Monte Carlo approximations to the HJB formulation will be implemented, and parameter robustness testing will be carried out if the core framework converges efficiently.

Phase 4 — Synthesis and Extended Investigation (Weeks 7–8). We will synthesize results from Phases 1–3 into a comprehensive technical report with figures, analysis, and protocol comparisons. Two exploratory directions will be pursued pending progress: first, we will investigate how model parameters can be inferred from population trajectory data and how estimation uncertainty propagates into control performance; second, as a speculative extension, we will explore whether actor-critic reinforcement learning [5] offers a tractable alternative for learning adaptive dosing policies directly from simulated trajectories in the time-varying stochas-

tic setting. These components develop methodology for future data-driven and learning-based deployment.

Budget, Duration and Facilities

Budget

The budget supports computational infrastructure and specialized research software essential for stochastic modeling and Monte Carlo simulations. Hardware investments (SSD, backup) provide permanent departmental assets for data-intensive computation. Research software subscriptions cover literature analysis, code development, mathematical verification, and specialized databases necessary for advanced research.

Item	Cost (Rs.)
SSD / external storage for simulations	12,000
RAM upgradation for computational work	7,000
Advanced literature & research tools	10,000
Cloud computing & HPC credits	2,000
Books, documentation & resources	3,000
Contingency & miscellaneous	6,000
TOTAL	50,000

Facilities

- IIT Delhi HPC cluster for large-scale parallel Monte Carlo simulations.
- Personal computing hardware (with above upgrades) for local development, testing, and intermediate-scale simulation.
- IIT Delhi library and journal subscriptions for access to research literature.

Duration

The project spans eight weeks during the summer. Core objectives (Phases 1–3) are designed for weeks 1–6, with weeks 7–8 reserved for synthesis, report writing, and exploratory extensions pending progress.

Acknowledgements

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